COMPUTERS & GRAPHICS | JOURNAL STUDY

Timestep-Conditioned Adapter Scaling for Multi-view Diffusion Texture Generation

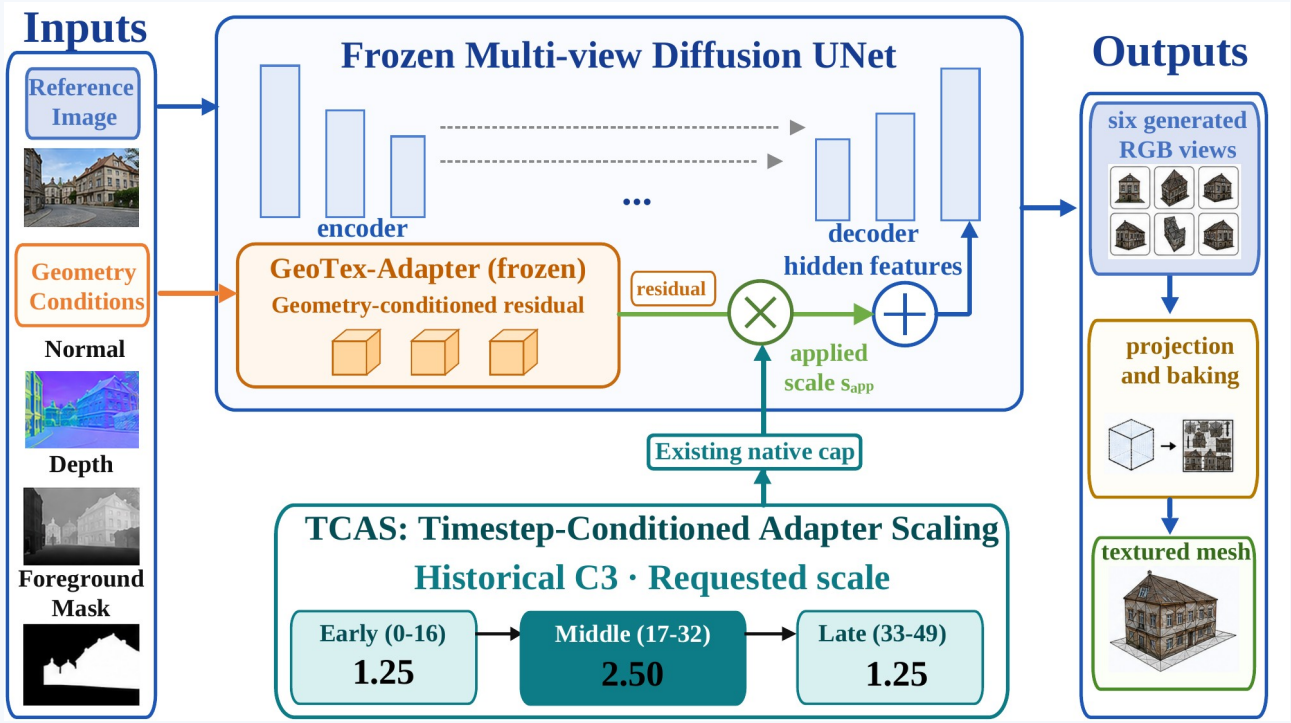
Control allocation and baked appearance

A controlled evaluation of an existing geometry-residual interface,
from applied scale to UV-textured assets viewed from unseen cameras.

CAD/Graphics 2026 Paper 75 | Revised journal evaluation

The graphics result is the textured asset, not only the generated view

- Generated RGB views are intermediate outputs in a 3D texturing pipeline.
- Projection, UV coverage, hole filling, and resampling affect the final appearance.
- We evaluate baked meshes from cameras withheld from texture generation and baking.



Study path: executed control → generated views → native-UV bake → unseen-view appearance

Follow the executed control, not the schedule label

Historical C3 request across 50 denoising steps

EARLY · 0–16	MIDDLE · 17–32	LATE · 33–49
1.25	2.50	1.25

Fresh C: deep and middle groups retain the requested profile; shallow requests are capped at 0.80 in every window.

THREE DISTINCT QUANTITIES

Requested scale → cap-adjusted applied scale → state-dependent residual correction

Equal scalar sums do not imply equal accumulated residual dose.

Strong controls revise the interpretation of C3

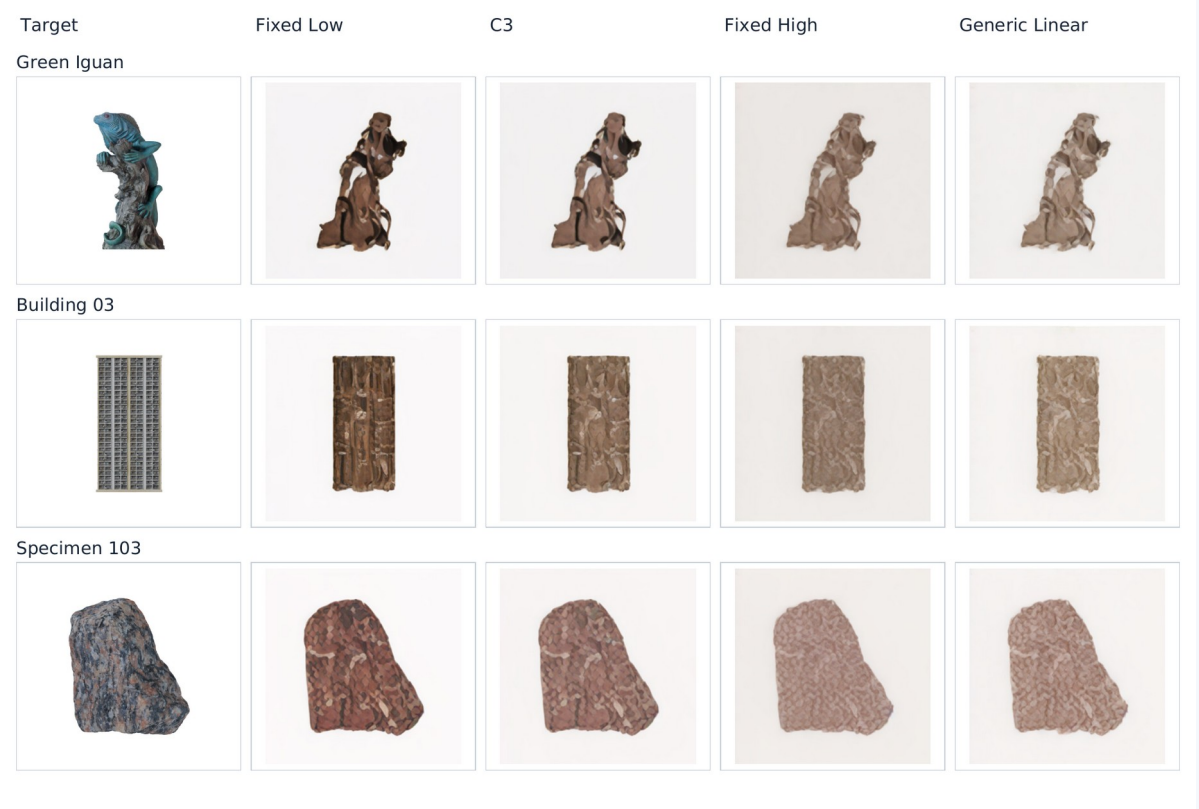
Fresh C · 300 held-out objects

Condition	FG-PSNR ↑	FG-LPIPS ↓
Fixed Low	11.31	0.2051
C3 / Middle	**11.83**	**0.1998**
Fixed High	12.58	0.1891
Generic Linear	12.42	0.1862

C3 − Fixed Low: +0.522 dB FG-PSNR, while deep/middle applied sums rise from 62.5 to 82.5.

Fixed High and Generic Linear have better means than C3 on all four image endpoints.

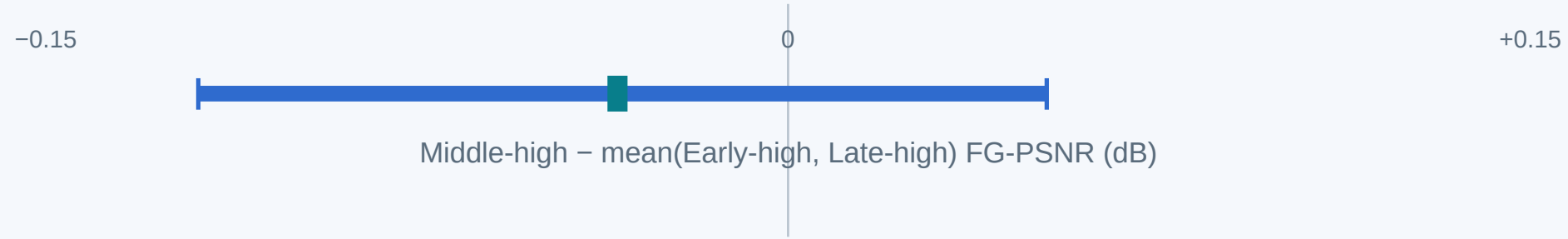
Same target, five conditions, three display objects



Examples illustrate condition differences; they do not estimate population performance.

Matched applied-scale budgets detect no middle-window advantage

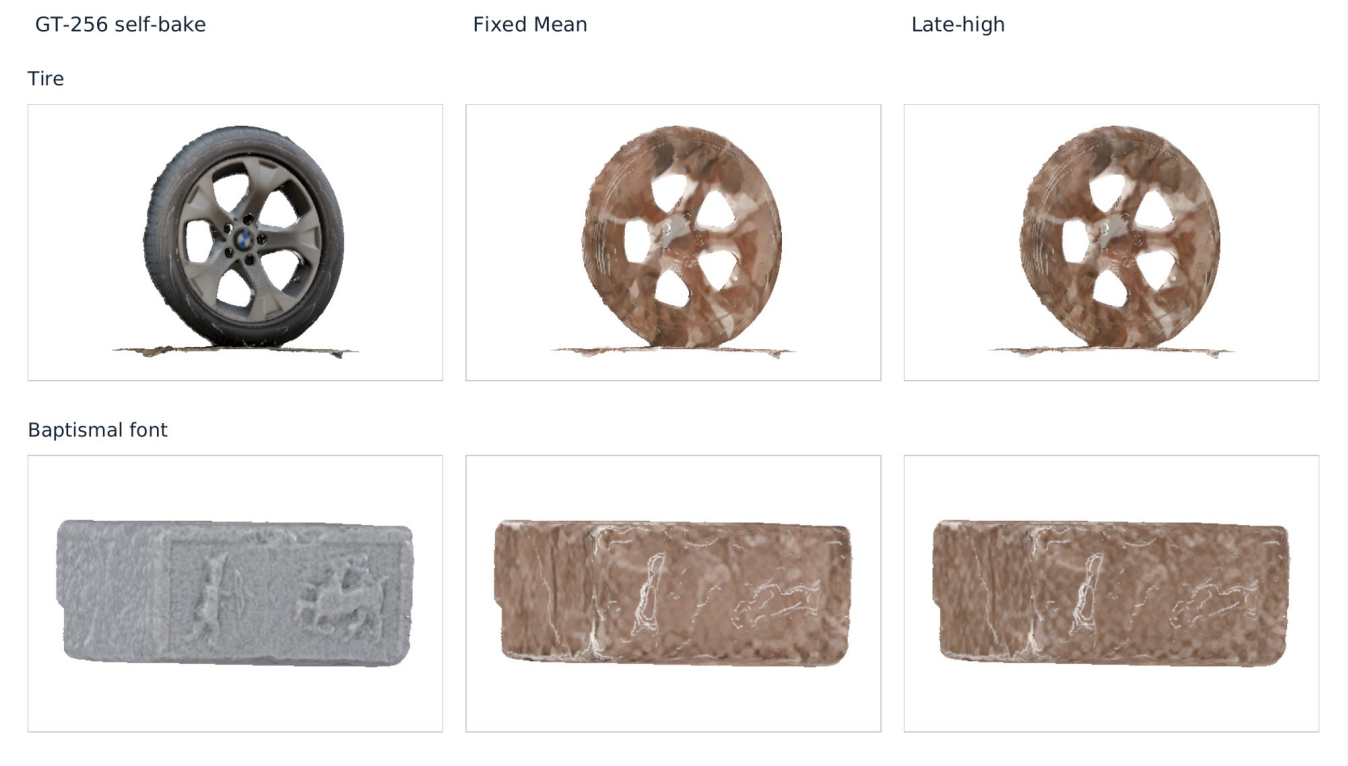
Fresh C · N = 300 · per-group applied-scale sums = 82.5 / 82.5 / 40



−0.0339 dB 95% interval [−0.1172, +0.0514] p = 0.4324

The specified primary does not support a middle-window advantage; it also does not establish equivalence. Retrospective early/late pairwise contrasts differ in opposite directions, showing placement sensitivity without a preferred schedule.

A small baked-view effect does not make C3 the winner



GEOTEX BAKE PILOT · N = 10

Late-high − Fixed Mean unseen-view FG-LPIPS −0.00265 [−0.00414, −0.00137]

All ten objects favored Late-high. The change closes 3.7% of the mean reference self-bake gap.

Late-high is distinct from C3. The separate GSO bake (N = 20) did not detect the same benefit: +0.000084 LPIPS, interval [−0.000103, +0.000286].

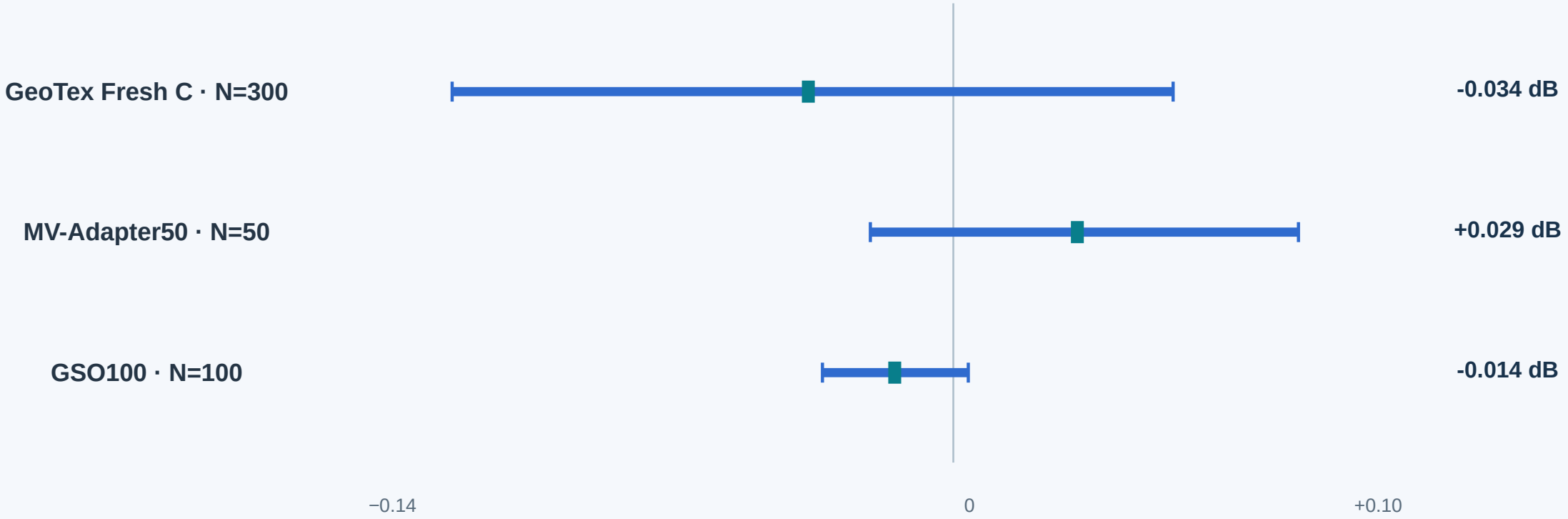
Compatibility-selected pilot; substantial reference-appearance error remains.

Image-space gains do not imply faithful reference appearance



Reference-color and texture discrepancies remain visible. Unlit base-color renders do not establish PBR or material fidelity.

Native-interface tests bound transfer across implementations



All three intervals include zero. The studies use separate cohorts and study-specific contrasts; GSO100 shares the MV-Adapter interface and is not an independent backbone replication.

The contribution is an evidence chain for deployed geometry control

01 IMPLEMENTATION

Separate requested scales, cap-adjusted applied scales, and state-dependent residual corrections.

02 CONTROLLED COMPARISONS

Compare C3 with strong fixed and linear controls and a matched applied-scale timing test.

03 ASSET-LEVEL EVALUATION

Connect image-space results to native-UV bakes from unseen cameras and report transfer boundaries.

The evidence supports reporting the executed interface, applied control, cohort, and asset-level response. It does not identify a universally preferred schedule or establish faithful material reproduction.

CONCLUSION | LIMITED PUBLIC DATA

No universal timing winner.
The baked asset remains the graphics endpoint.

C3 improves over Fixed Low, but stronger controls lead on mean image endpoints. The matched-budget primary detects no middle-window advantage. A small GeoTex bake benefit does not recur in the separate GSO protocol.

ZENODO · VERSION 1.0.1 · LIMITED DERIVED DATA

DOI: 10.5281/zenodo.23383807

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Thank you

https://doi.org/10.5281/zenodo.23383807